Sujal Lohar

Full-Stack Software Engineer & DevSecOps Architect

Ahmedabad, Gujarat, India
sujapanchal17@gmail.com • +91-7665351906
github.com/sujallohar •
linkedin.com/in/sujallohar

SUMMARY

Software Engineer and DevSecOps Architect with core competencies in distributed microservices (Java 21, Spring Boot 3.2, gRPC, PostgreSQL, Redis), container security, and cryptographic audit ledgers. Experienced in designing Saga choreography state machines, multi-stage non-root Docker deployments, and automated CI/CD security quality gates.

EDUCATION

Institute of Technology, Nirma University

Aug 2023 – May 2027

Bachelor of Technology (B.Tech) in Computer Science and Engineering (Minor in Cybersecurity) — ***8.44 / 10.0 CGPA***

- Core Coursework: Distributed Systems, Operating Systems, Computer Networks, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming (Java/C++), Cryptography & Network Security, Cloud Computing.
- Recipient: Ministry of External Affairs SPDC Full Scholarship (2023 - 2027).
- Competitive Programming: Codeforces Specialist (Rating 1400+), solve algorithmic graph and dynamic programming problems daily.

Swami Vivekanand Govt. Model School

Jul 2022 – May 2023

Senior Secondary Certificate (Class XII) in Science Stream (PCM) — ***95.8%***

- Awarded Rajasthan State Scholar Student Laptop Award for academic excellence in district rankings.

Swami Vivekanand Govt. Model School

Jul 2020 – May 2021

Secondary School Certificate (Class X) in General Curriculum — ***96.2%***

- Top rank in secondary district board examinations.

EXPERIENCE

Undergraduate Research Fellow | Institute of Technology, Nirma University

Aug 2024 – Present

Ahmedabad, Gujarat, India

- Engineered decentralized federated learning pipelines using FedProx across 40,000+ MIMIC-III ICU records.
- Attained 96.4% ICU hazard classification accuracy with multi-hop attention rollout for clinical interpretability.
- Co-authored 5 peer-reviewed papers indexed in IEEE Xplore and presented at international conferences.

Automation & Web Developer Intern | T.I.M.E. (Triumphant Institute of Management Education)

Jun 2025 – Jul 2025

Ahmedabad, Gujarat, India

- Constructed containerized Python dispatch service using Twilio WhatsApp API and MongoDB, increasing delivery speed by 30%.
- Eliminated 10+ hours per week of manual operational overhead for administrative staff.

Web Development Intern | The Akshaya Patra Foundation

Jun 2024 – Jul 2024

Ahmedabad, Gujarat, India

- Engineered operational web app on Node.js, Express, and MySQL replacing paper logs across 2 industrial kitchens.
- Reduced reporting latency and saved 10+ hours per week in manual kitchen reconciliation.

Machine Learning Intern | Suven Consultants & Technology Pvt. Ltd.

Jul 2024 – Aug 2024

Remote / Mumbai, India

- Built feature engineering pipelines in Python and Scikit-Learn for tabular data classification.
- Packaged model inference workers into Flask REST APIs deployed on cloud endpoints.

Selected Delegate (HPAIR Tokyo) | Harvard Project for Asian and International Relations

Aug 2025 – Oct 2026

Tokyo, Japan

- Selected among international applicants (sub-10% acceptance rate) to represent India and Nirma University at Tokyo summit.
- Collaborated on technology policy and public infrastructure security frameworks.

Social Media & Communications Manager | Rotaract Club of Ahmedabad Emerald Jul 2025 – Present
Ahmedabad, Gujarat, India

- Direct technical and digital engagement campaigns across community welfare, youth development, and social action programs.
- Coordinated cross-functional outreach with executive board members to expand community participation.

Junior Manager, OGTA | AIESEC in Ahmedabad Aug 2024 – Feb 2025
Ahmedabad, Gujarat, India

- Assessed technical student qualifications and matched candidates with international engineering internships.
- Conducted interview prep, technical resume reviews, and visa guidance for outgoing candidates.

SELECTED ENGINEERING PROJECTS

GovAsset Nexus: Polyglot Microservices Platform Oct 2026
Stack: Java 21 • Spring Boot 3.2 • Redis 7.2 • gRPC • PostgreSQL • MongoDB • Kafka • Docker • Kubernetes • Istio

- Architected 5 polyglot microservices on Java 21 / Spring Boot 3.2 and Node.js using gRPC and Kafka.
- Engineered distributed Saga choreography with PostgreSQL transactional outbox for zero-loss recovery.
- Deployed to Kubernetes with Istio service mesh and Redis 7.2 sliding-window rate limiting.

GovAsset 360: Cryptographic WORM & AppSec Gateway Sep 2026
Stack: Node.js • TypeScript • Web Crypto SHA-256 • Firestore WORM • Docker • Trivy • GitLeaks • GitHub Actions • Zod

- Engineered SHA-256 cryptographic hash-chained audit engine with WORM append-only enforcement.
- Constructed hardened AppSec Express gateway with Zod schema validation and sliding-window rate limiters.
- Built multi-stage non-root Docker builds verified via Trivy and GitLeaks CI/CD security gates.

Federated Clinical Predictor & XAI Engine Aug 2025
Stack: Python 3.11 • PyTorch • FedAvg / FedProx • TabTransformer • MIMIC-III Dataset • Scikit-Learn • Flask • Docker

- Formulated decentralized federated learning pipeline using FedProx on 40,000+ MIMIC-III ICU records.
- Attained 96.4% predictive accuracy with multi-hop attention rollout for clinical feature interpretability.
- Co-authored research paper published in IEEE Xplore and presented at international conferences.

Edge IoT Health Telemetry Pipeline May 2025
Stack: C++ • ESP32 • MQTT (TLS 1.3) • InfluxDB • Grafana • Docker • Python • WebSocket

- Programmed ESP32 C++ firmware for 100Hz physiological biometric sensor sampling.
- Engineered TLS 1.3 MQTT telemetry ingestion pipeline into InfluxDB and Grafana dashboards.

FocusFlow: Client-Side Computer Vision Telemetry Jul 2025
Stack: React 19 • TypeScript • MediaPipe Wasm • Web Audio API • Tailwind CSS • Vite

- Engineered client-side computer vision pipeline processing 478 facial landmarks at 30 FPS in WebAssembly.
- Guaranteed 100% video stream privacy by eliminating cloud telemetry.

NASA Near-Earth Object Orbital Risk Engine Oct 2024
Stack: Three.js • WebGL • TypeScript • Python FastAPI • NASA JPL API • Docker

- Constructed Three.js 3D orbital simulator processing 30,000+ near-Earth asteroids from NASA JPL APIs.
- Integrated Python hazard classifier predicting close-approach risk factors.

RESEARCH PUBLICATIONS

"Federated Transformer-Based Learning for Predicting ICU Events with Multi-hop XAI Explanation Models" — IEEE Conference (Published in IEEE Xplore) (IEEE, 2025) [Published]

"Privacy-Preserving Federated Healthcare Intelligence via Blockchain and Differential Privacy" — COMSYS 2025: 6th International Conference on Frontiers in Computing and Systems (Springer / IEEE (Warsaw, Poland), 2025) [Accepted]

"Explainable Deep Learning Framework for Intensive Care Unit Anomaly Forecasting" — COMSYS 2025: 6th International Conference on Frontiers in Computing and Systems (Springer / IEEE (Warsaw, Poland), 2025) [Accepted]

"Multi-Hop Attention Rollout in Clinical Event Sequences" — COMSYS 2025: 6th International Conference on Frontiers in Computing and Systems (Springer / IEEE (Warsaw, Poland), 2025) [Accepted]

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript / TypeScript, Java, SQL, Bash / Shell

Cloud, DevOps & DevSecOps: Docker, Kubernetes, GitHub Actions, DevSecOps (Trivy / GitLeaks), AWS & OCI, Linux System Administration

Backend & Distributed Systems: Spring Boot 3.2, Node.js & Express, NestJS, gRPC & Protocol Buffers, Kafka & Redis

Security, Cryptography & Observability: Cryptographic SHA-256 Ledger, Prometheus & Grafana, Falco Runtime Detection, Active Defense Honeypots

Databases & Storage: PostgreSQL, MongoDB, Redis, Firebase Firestore

Machine Learning & Federated Systems: Federated Learning (FedAvg / FedProx), Explainable AI (XAI), MediaPipe Computer Vision, Scikit-Learn, NumPy & Pandas

CERTIFICATIONS & HONORS

Oracle Cloud Infrastructure 2024 DevOps Professional (Oracle Cloud Infrastructure, Oct 2024)

DevOps Engineering Job Simulation (Macquarie Group (via Forage), May 2024)

NASA Space Apps Challenge Global Nominee (NASA - National Aeronautics and Space Administration, Oct 2024)

Selected Delegate - HPAIR Harvard Conference (Harvard College Project for Asian and International Relations (Tokyo, Japan))

UG Research Fellowship Grant (10,000 INR) (UG Research Club, Nirma University)

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional (Oracle Cloud Infrastructure, Aug 2024)

Cybersecurity Fundamentals (Palo Alto Networks Cybersecurity Academy, 2024)

Top 8 Sanchar Mitras State Recognition (Department of Telecommunications (DoT), Government of India - Gujarat LSA)

Scholarship Programme for Diaspora Children (SPDC) (Ministry of External Affairs, Government of India)

Top 100 Finalist (Asia-Pacific) (Global Goals Summit 2025)